

# Chapter: Modifiability Graph Diagnostics and Tier-Nested Operator Architecture

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*A dissertation-chapter style exposition integrating HTQA and RED/DED*

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## Chapter abstract

This chapter reformulates the modifiability-graph diagnostic for nominal architecture as a reproducible method suitable for cross-linguistic deployment. The core proposal is a tier-nested architecture: LOW (argumental primitives such as reciprocals, nominal WH heads, and free relatives) is composed inside MIDDLE (packaging/repair mechanisms: ellipsis, pronounization, prop-noun/classifier insertion, and typed coercion), which is composed inside HIGH (operator control: D/Q/Deg/Prox/Neg; HTQA entry mode; closure-sensitivity). A strict notion of head-modification defines directed graphs per tier; graph partitions (isolates and SCCs) are linked to architectural role, yielding falsifiable stacking and repair predictions. We instantiate Tier-1 reference-attested graphs (nodes/edges/density; nontrivial SCC in MIDDLE), and we fully derive a  $\kappa$ -licensed group-distributive case showing where structured-domain  $\kappa$  is required.

## Roadmap

Section 1 motivates the method and the tier stack. Section 2 introduces HTQA and RED/DED as HIGH-tier controllers and wrapper procedures. Section 3 defines strict head-modification and the modifiability graph, with an explicit decision rubric. Section 4 builds a closure-sensitivity feature system and derives a stacking grammar. Section 5 instantiates tier graphs and reports isolates/SCCs/density. Section 6 develops cross-linguistic diagnostic mini-sets (English, German, Mandarin, Turkish) with uniform tests, glossing, and judgment labels. Section 7 executes a  $\kappa$ -licensed group-distributive derivation. Sections 8-10 present distinguishing predictions plus computational and neuroscience implications. Appendices provide coding rubrics and results templates.

## Key terms (quick reference)

| Term                     | Definition (chapter usage)                                                                   |
|--------------------------|----------------------------------------------------------------------------------------------|
| Strict head-modification | Direct head targeting with constituency; no category shift; excludes phrase-level scope.     |
| Modifiability graph      | Directed graph over a tier inventory: $X \rightarrow Y$ iff $X$ strictly head-modifies $Y$ . |

|                              |                                                                                                                                |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Tier nesting                 | Containment: $LOW \subset MIDDLE \subset HIGH$ ; wrapper RED/DED( $H[M[L]]$ ).                                                 |
| HTQA entry mode (TET)        | Preclosure ALL (Mode 1), remainder-lattice entry (Mode 2), postclosure traversal EACH/EVERY/NO (Mode 3).                       |
| Repair suppliers             | ELL (NP-ellipsis), PRON (lexical DP replacement), REPAIR (prop-noun/classifier insertion), COERCE (typed predicate formation). |
| $\kappa$ (structured domain) | Contextual cover/partition parameter needed for group distributives (e.g., 'every three N').                                   |

## 1. Why a modifiability graph?

Traditional operator diagnostics in DP and generalized-quantifier (GQ) work often aggregate many tests (stacking bans, substitution, scope behavior, repair patterns) without a single mathematical object that integrates them. The modifiability-graph proposal converts one recurrent intuition into an explicit computable structure: within a fixed inventory (Level-k) we draw a directed edge only when strict head-modification is attested. This yields isolates, SCCs, and partial orders that can be compared across languages, registers, and annotation regimes. The resulting partitions are not definitional: they can be wrong, and thus empirically falsified.

The dissertation-level advantage is methodological: the graph object supports explicit reporting (nodes, edges, density), and creates a bridge to computational induction (graph learning from parsed corpora) and to experimental design (predicting where the parser must recruit repair).

### 1.1 Tier nesting as the chapter's organizing hypothesis

We use a three-tier containment stack: LOW content is packaged by MIDDLE and only then enters HIGH operator control. RED and DED wrap the entire object as paired inference/prediction procedures.

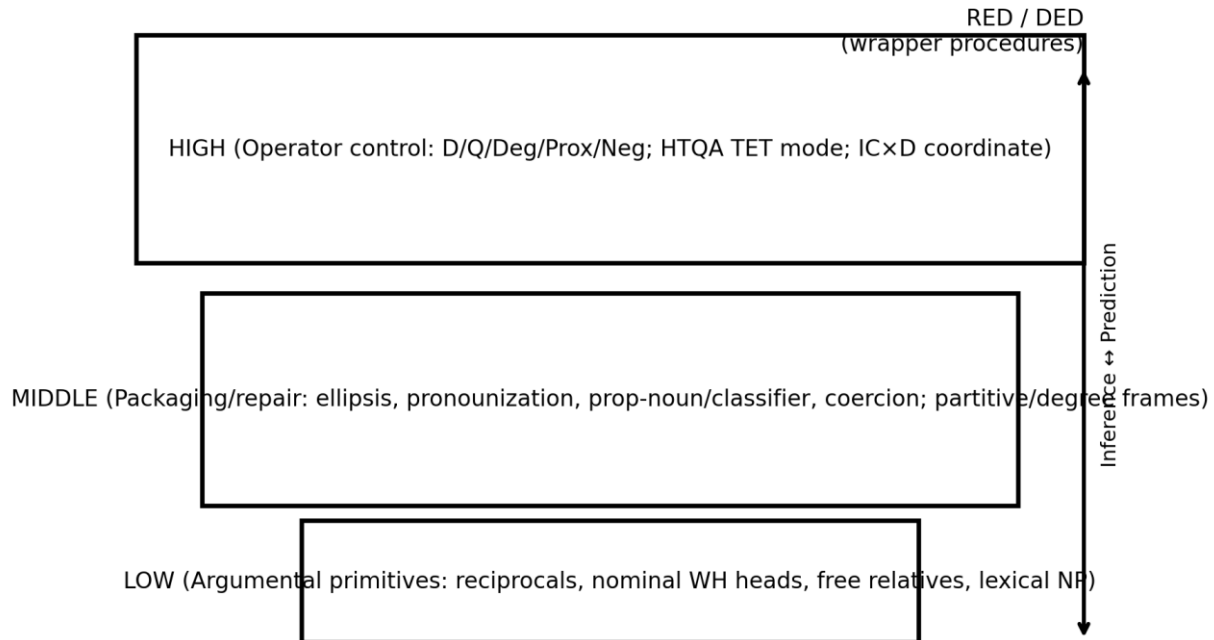


Figure 1. Tier nesting:  $LOW \subset MIDDLE \subset HIGH$ , wrapped by RED/DED procedures.

## 1.2 Method overview

The chapter's workflow is a diagnostic pipeline: classify items into tier inventories, apply strictness filters, update tier graphs, infer HTQA entry mode and repair type, then derive stacking and repair predictions.

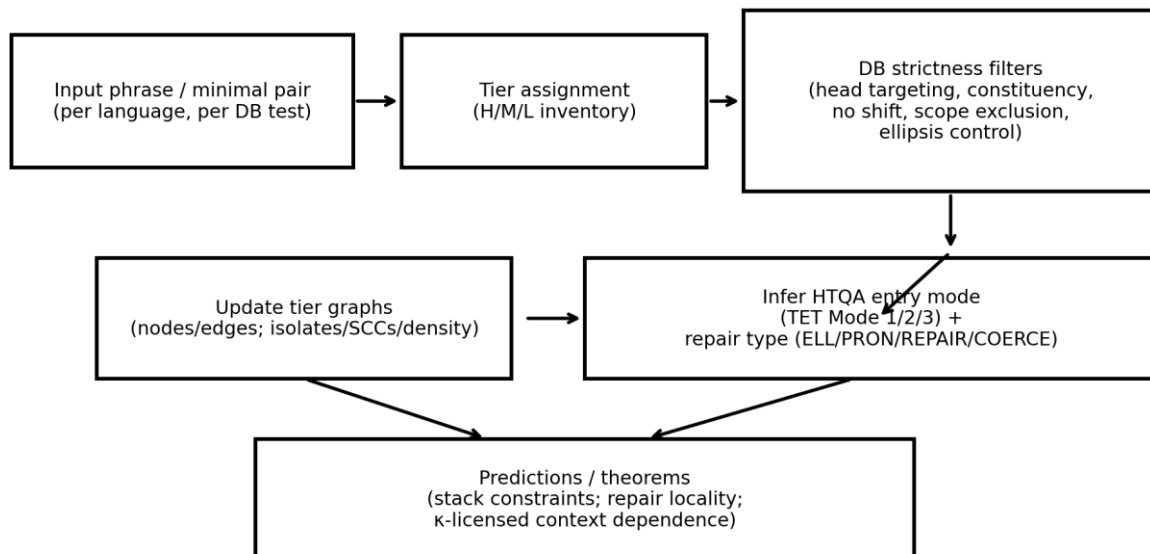


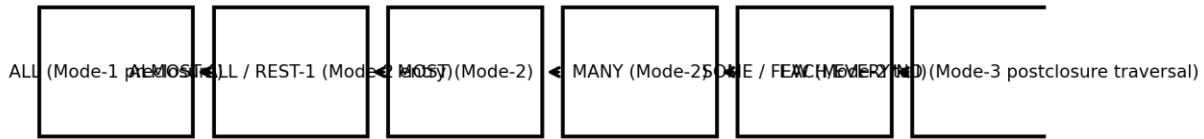
Figure 2. Diagnostic pipeline from minimal pairs to tier graphs and predictions.

## 2. HTQA and RED/DED in the tier stack

HTQA is treated here as a HIGH-tier controller: it supplies entry-mode distinctions (TET) and a remainder-lattice generated by iterating unit extraction  $\langle 1, \text{REST} \rangle$ , yielding remainder-banded localities  $\text{IC}(\text{REST})$ . RED/DED operate as wrapper procedures: RED identifies latent architectural commitments from observed distributions; DED generates derivable predictions (including repair-locality constraints) from the architecture.

### 2.1 HTQA entry modes as tier-external control

For readers new to HTQA, the key point is that 'ALL' and post-closure items such as 'each/every/no' are not merely different lexical labels: they enter composition by different modes. The entry mode controls what counts as an addressable unit for traversal, which becomes crucial for  $\kappa$ -licensed group readings.



Interpretation: Modes differ in entry object and addressability; Mode-3 requires explicit unit traversal (often distributive).

Figure 2a. HTQA ladder schematic: entry modes differ in addressability and traversal requirements.

The practical interpretation for this chapter is simple: (i) HIGH determines which kinds of objects are addressable for quantification (atoms vs groups vs remainder bands), (ii) MIDDLE supplies missing  $\langle e, t \rangle$  material or coercions when HIGH selection fails, and (iii) LOW provides the minimal argumental heads and nominal predicates that must be visible to closure.

## 3. Definitions: strict head-modification and tier graphs

### 3.1 Strict head-modification (edge criterion)

Let  $X$  and  $Y$  be POS items or POS-classes within a fixed tier inventory. We add an edge  $X \rightarrow Y$  if and only if there exists an attested configuration satisfying all of: (i) head targeting ( $X$  attaches to  $Y$  as a head, not merely to a phrase containing  $Y$ ), (ii) constituency ( $[X Y]$  is a constituent excluding the rest of the phrase), (iii) no category shift (no quotation or metalinguistic mention), and (iv) scope exclusion (if  $X$  scopes over a phrase containing  $Y$  but does not target  $Y$  as head, no edge is added).

### 3.2 Ellipsis/pro-drop control

A bare form counts as true headability only if the language disallows NP-ellipsis/pro-drop in the relevant configuration (or independent diagnostics show no silent NP). Otherwise the configuration is classified as ellipsis-supported and does not justify adding a head-modification edge.

Required computed statistics per tier are: (i)  $|V|$  (#nodes) and  $|E|$  (#edges); (ii) edge density  $\rho = |E| / (|V| \cdot (|V|-1))$  for  $|V| > 1$ ; (iii) the isolate set  $\text{Iso}(G) = \{v \in V : \text{indeg}(v)=0 \wedge \text{outdeg}(v)=0\}$ ; (iv) strongly connected components  $\text{SCC}(G)$  with  $\text{size} > 1$ ; and (v) optional degree distributions (indegree/outdegree histograms) when comparing corpora/genres. The core interpretive use is the contrast between isolate-dominated tiers (predicted for HIGH under strict criterion) and connected tiers (expected in MIDDLE via iterative packaging).

Interpretation rule-of-thumb (to be tested cross-linguistically): HIGH isolates indicate that apparent 'modification' among operators is scope-level rather than head-level; MIDDLE SCCs indicate mutually iterative packaging heads (a nontriviality witness); LOW is typically sparse, with edges limited to tightly constrained modifier constructions (e.g., relative-modifier  $\rightarrow$  nominal head) and lexical compounding when treated as head-level.

In dissertation reporting, this subsection also functions as the protocol's audit trail: when a reviewer questions whether the partition results are 'definitional,' the SCC and density computations provide a falsifiable target. If a purported HIGH-tier operator exhibits robust head-modification (non-isolate behavior) under the strict rubric, the architecture must be revised rather than re-labeled.

### 3.3 Graph objects and reporting

This subsection specifies the exact reporting obligations of the modifiability- graph method so that it is auditable and reproducible. For each tier  $k$  we construct a directed graph  $G_k = \langle V_k, E_k \rangle$  where  $V_k$  is the tier inventory (POS items/classes) and  $E_k$  contains only strict head- modification edges  $X \rightarrow Y$  (as defined in §3.1–§3.2). We then compute a fixed set of graph summaries that must be reported for every language and every evidence tier (reference- attested, corpus- attested, or elicitation).

Required computed summaries per tier are:

- (i)  $|V|$  (#nodes) and  $|E|$  (#edges).
- (ii) Edge density  $\rho = |E| / (|V| \cdot (|V|-1))$  for  $|V| > 1$ .
- (iii) Isolate set  $\text{Iso}(G) = \{v \in V : \text{indeg}(v)=0 \wedge \text{outdeg}(v)=0\}$ .
- (iv) Strongly connected components  $\text{SCC}(G)$  with  $\text{size} > 1$  (mutual reachability).
- (v) Optional degree distributions (indegree/outdegree) when comparing corpora or genres.

Interpretation rule- of- thumb (to be tested cross- linguistically): HIGH isolates indicate that apparent “operator modification” is scope- level rather than head- level; MIDDLE SCCs indicate mutually iterative packaging heads (a nontriviality witness); LOW is typically sparse, with edges limited to tightly constrained modifier constructions (e.g., relative- modifier  $\rightarrow$  nominal head) and lexical compounding when treated as head- level.

| Quantity       | Definition                                | Why it matters (diagnostic use)                                               |
|----------------|-------------------------------------------|-------------------------------------------------------------------------------|
| $ V ,  E $     | Nodes and strict head- modification edges | Baseline complexity; enables cross- language comparability.                   |
| $\rho$         | Density = $ E /( V ( V -1))$              | Separates isolate tiers from packaging- connected tiers.                      |
| Iso(G)         | Nodes with indeg=outdeg=0                 | Identifies tier- internal “non- modifiable” heads under the strict criterion. |
| SCC(G)         | Mutual- reachability components (size>1)  | Nontriviality witness; expected primarily in MIDDLE via iterative packaging.  |
| Degree profile | Indegree/outdegree distribution           | Helps separate sparse structural edges from corpus noise.                     |

Audit note: if a purported HIGH- tier operator exhibits robust head- modification (non- isolate behavior) under this strict rubric, the architecture must be revised (or the tier inventory re- assigned). Conversely, if MIDDLE does not yield any SCCs even after including attested packaging heads, that is evidence that the inventory is incomplete or the strictness filters are too strong.

### 3.4 Strict vs scope-level 'modification' (why the criterion matters)

A recurring confusion in reviews is to treat any operator taking semantic scope over a DP as 'modifying' the determiner head. Under the strict criterion, this does not count. For example, focus particles such as even/only/just can scope over an entire DP, but they do not head-modify THE. The strict graph therefore isolates THE even when DP-level particles occur. This is a substantive empirical commitment: if a language shows independent evidence that a particle attaches as a head to D (not to DP), the graph would gain a HIGH-tier edge and the architecture would have to be revised.

Operator feature lattice (schematic): permitted compositions are solid arrows; forbidden are dashed.

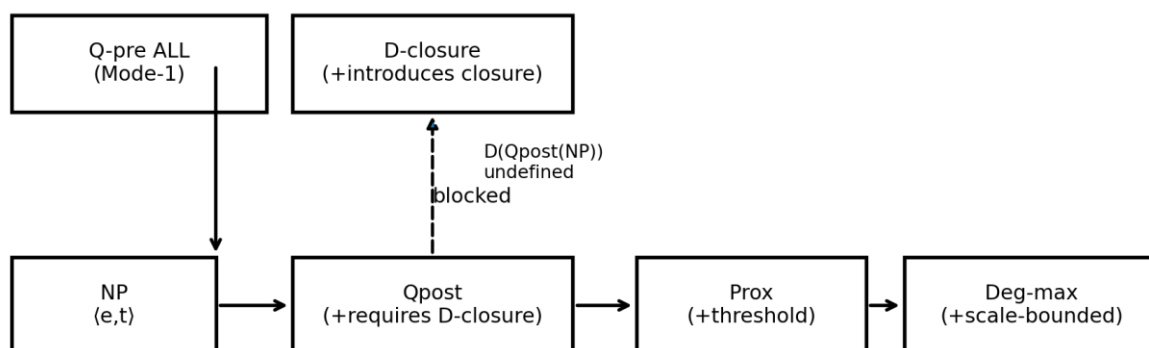


Figure 3a. Operator feature lattice (schematic): solid arrows are licensed compositions; dashed arrow indicates the predicted  $D(Qpost(NP))$  failure.

#### 4. Closure-sensitivity features and an allowed stacking grammar

We annotate operator-like POS with closure-sensitivity features:  $[\pm requires\ D-closure]$ ,  $[\pm introduces\ closure]$ ,  $[\pm scale-bounded]$ , and  $[\pm threshold]$ . These features summarize which operators need a D-closable NP stage, which operators perform closure, and which operators operate on bounded scales or thresholds.

| Type      | Examples                                            | Domain | requires D-closure | introduces closure | scale-bounded | threshold |
|-----------|-----------------------------------------------------|--------|--------------------|--------------------|---------------|-----------|
| Q-pre     | all                                                 | Q      | -                  | -                  | -             | -         |
| Q-post    | every, each, no                                     | Q      | +                  | -                  | -             | -         |
| D-closure | the, a/an, Ø                                        | D      | -                  | +                  | -             | -         |
| Deg-ampl  | very, extremely, so, too                            | Deg    | -                  | -                  | -             | -         |
| Deg-max   | absolutely, completely, entirely, totally, utterly, | Deg    | -                  | -                  | +             | -         |

|      |                       |      |   |   |   |   |
|------|-----------------------|------|---|---|---|---|
|      | perfectly             |      |   |   |   |   |
| Prox | almost,<br>nearly     | Prox | - | - | ± | + |
| Neg  | not, never,<br>barely | Neg  | - | - | - | - |

#### 4.1 Core stacking contrast

The feature system is designed to derive, rather than stipulate, central contrasts such as: almost every NP (licensed) versus \*the every NP (undefined under the closure pipeline). The explanation is typed: D-closure selects an NP predicate  $\langle e, t \rangle$ , while post-closure Q items return QP; hence  $D(Qpost(NP))$  is undefined. Proximity operators (almost/nearly) do not impose NP selection and can scope over QP.

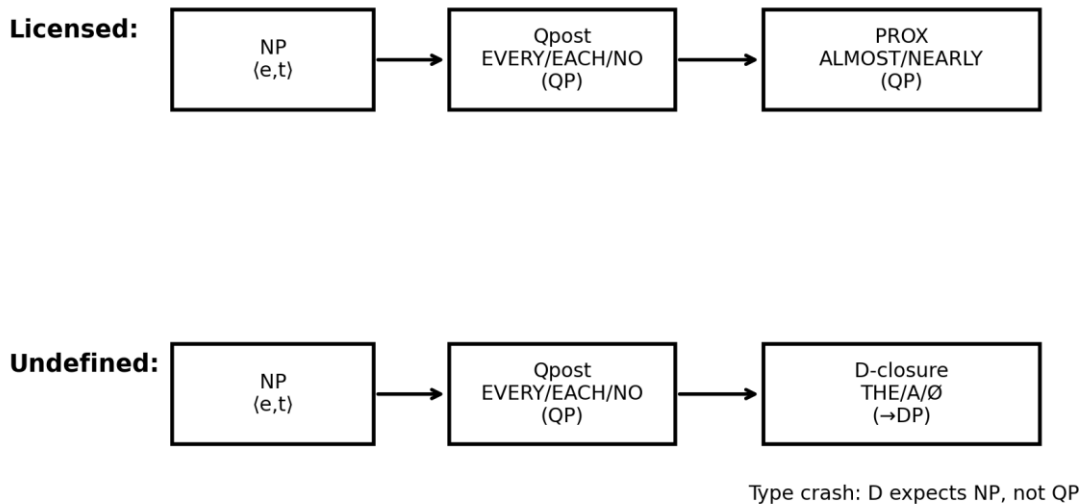
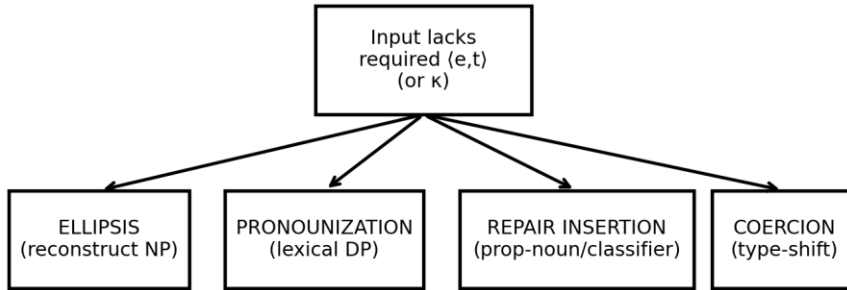


Figure 3. Derived stacking grammar: PROX(Qpost(NP)) is licensed, while D(Qpost(NP)) is undefined.





All are MIDDLE-tier suppliers; they satisfy the closure lemma (only these supply missing inputs without adding new primitives).

Figure 5a. MIDDLE-tier support strategies: Ellipsis, Pronounization, Repair insertion, and Coercion.

## 5. Instantiating tier graphs (Tier-1 reference-attested)

To demonstrate that the graph method is not definitional, we instantiate one graph per tier using conservative Tier-1 coding: we include only edges supported by standard descriptions or published examples, and we preserve the strict edge criterion. The MIDDLE tier contains a nontrivial SCC witness, while the HIGH tier remains isolate-dominated.

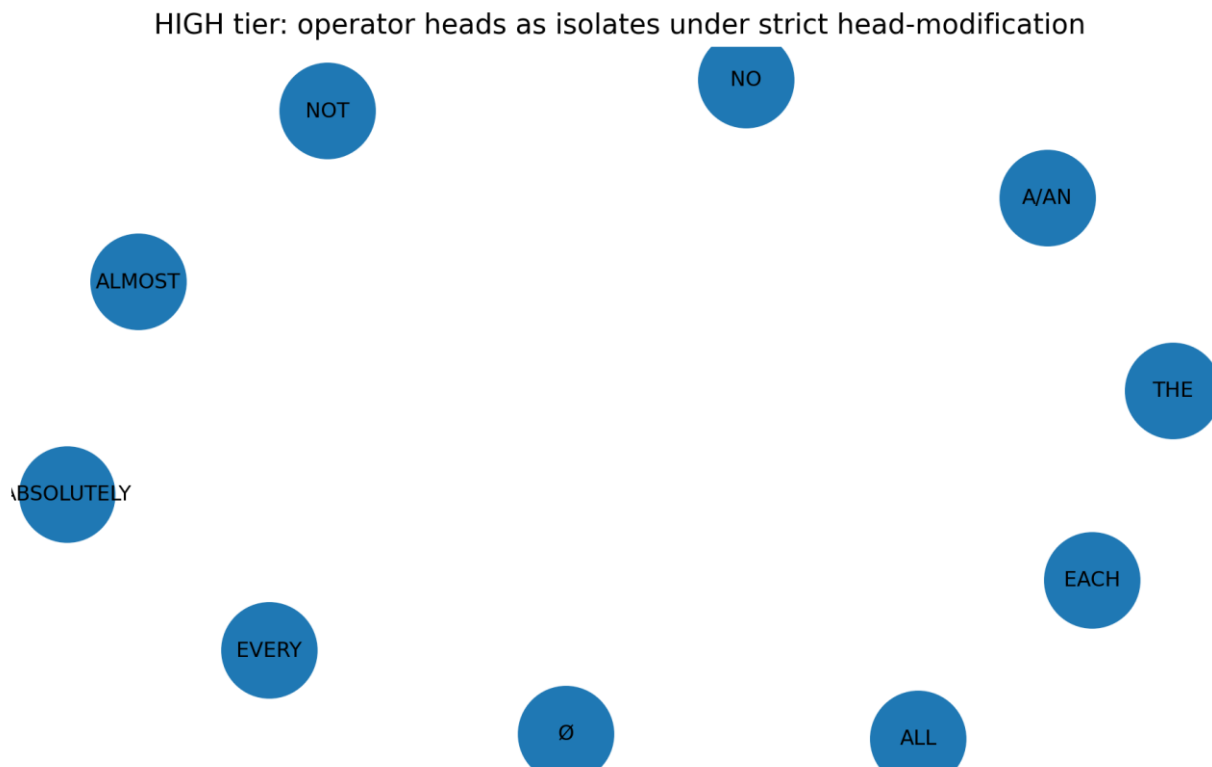


Figure 4. HIGH-tier modifiability graph: operator heads behave as isolates under strict head-modification.

MIDDLE tier: packaging/repair modifiability graph (SCC witness persists)

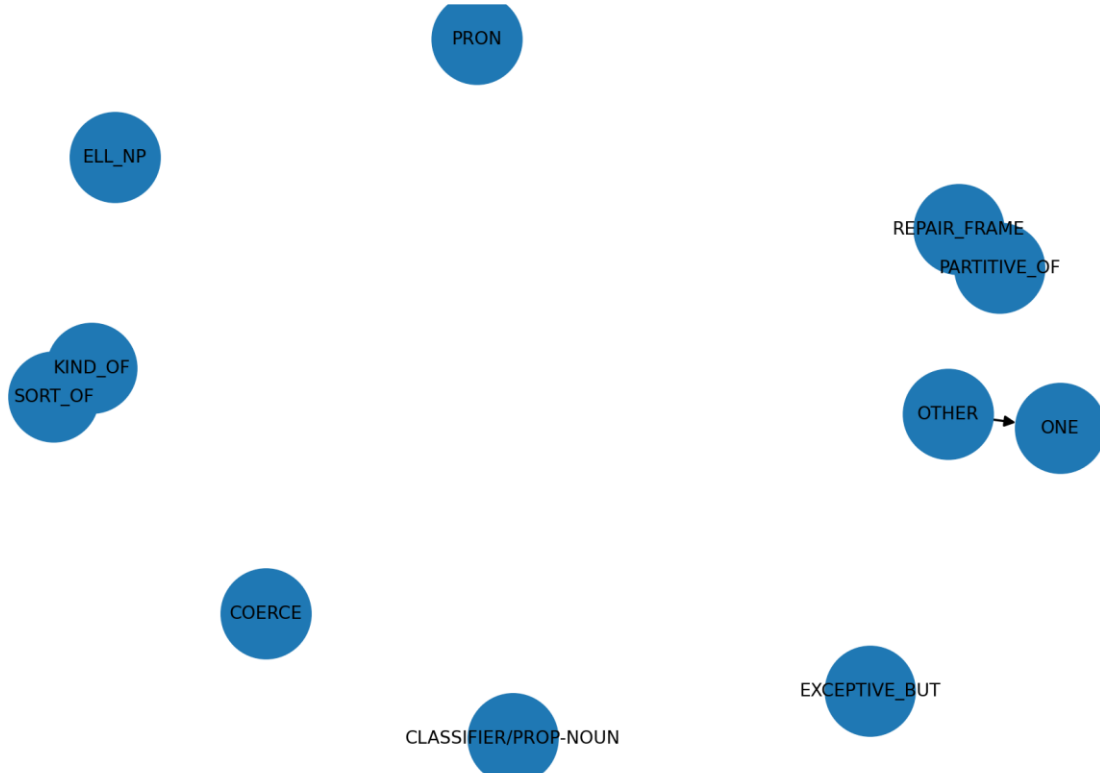


Figure 5. MIDDLE-tier modifiability graph: SCC witness ( $KIND\_OF \leftrightarrow SORT\_OF$ ) persists under strict criterion;  $OTHER \rightarrow ONE$  edge illustrates prop-noun packaging.

LOW tier: argumental primitives + restricted modifier-clause edges

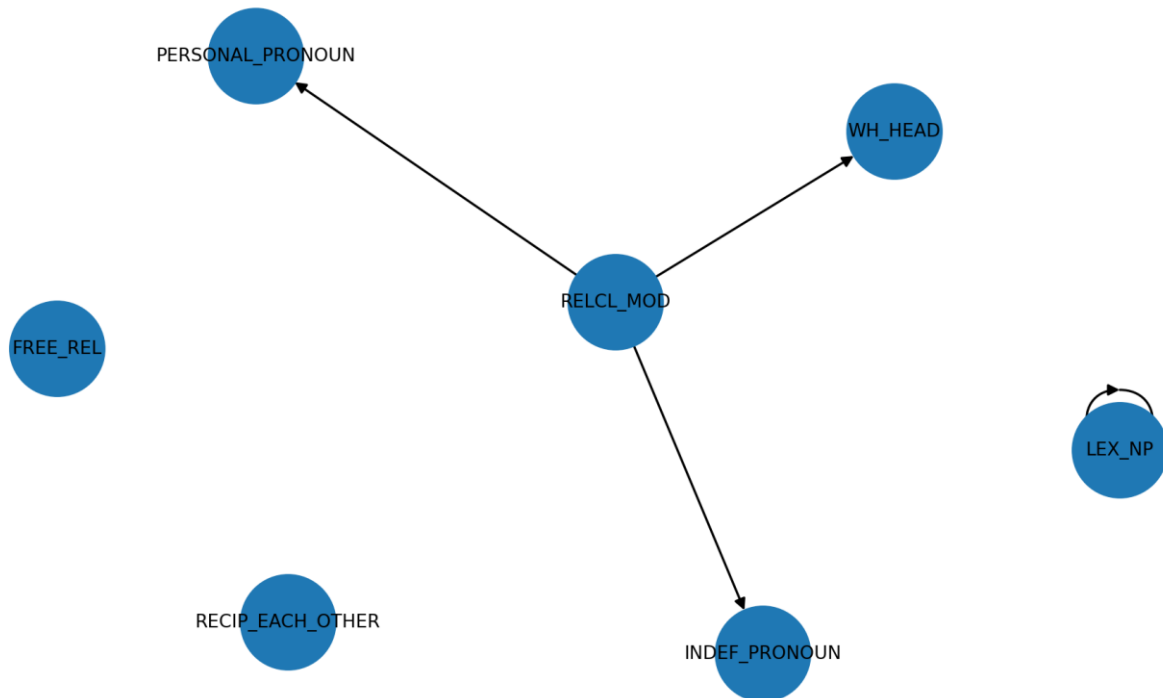


Figure 6. LOW-tier modifiability graph: sparse, dominated by restricted modifier-clause  $\rightarrow$  nominal-head relations.

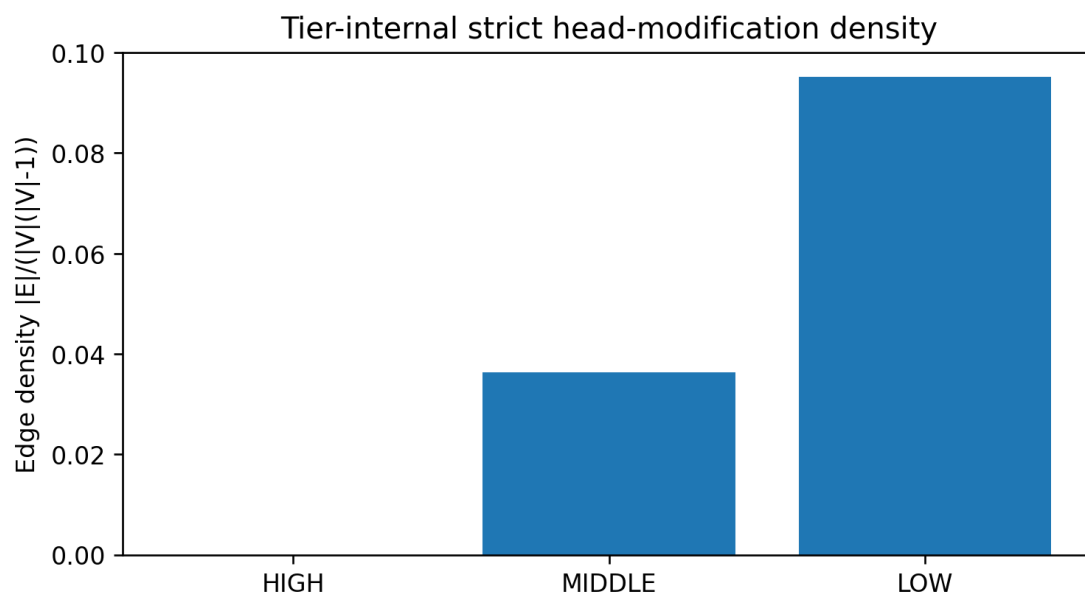


Figure 7. Tier-internal edge density under strict head-modification (instantiated graphs).

## 5.1 Graph statistics

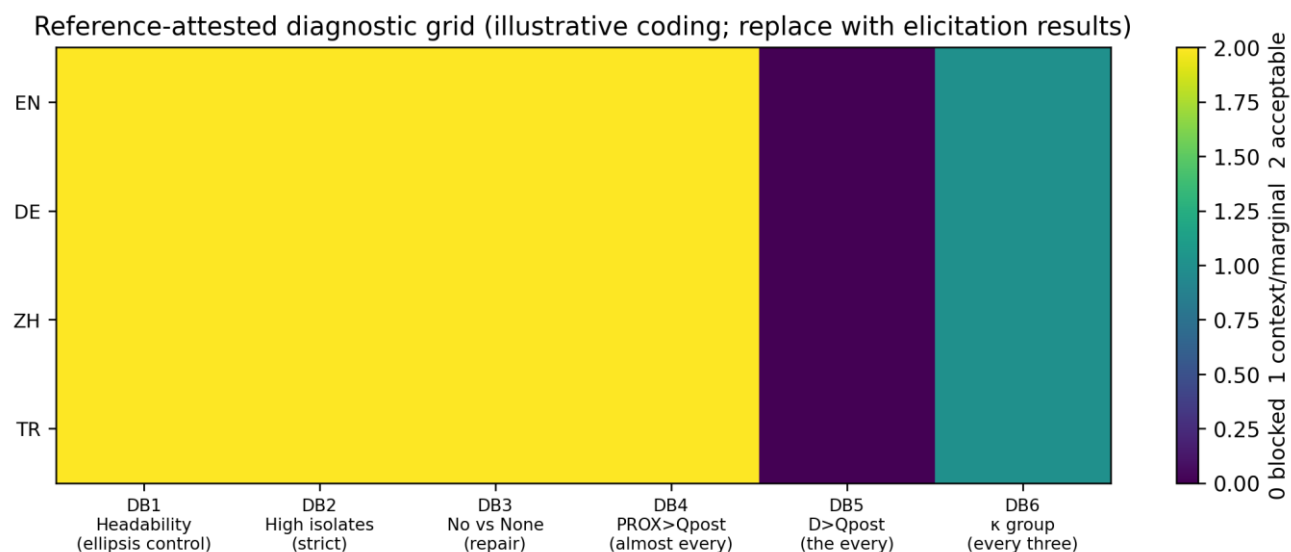


Figure 9. Reference-attested diagnostic grid (illustrative): replace with elicitation/corpus results as they become available.

| Tier   | #Nodes | #Edges | Density | Isolates                                                             | SCCs<br>(size>1) |
|--------|--------|--------|---------|----------------------------------------------------------------------|------------------|
| HIGH   | 10     | 0      | 0.000   | THE, A/AN, Ø, ALL,<br>EVERY, EACH, NO,<br>ALMOST,<br>ABSOLUTELY, NOT | —                |
| MIDDLE | 11     | 4      | 0.036   | ELL_NP, PRON,<br>CLASSIFIER/PROP-                                    | {SORT_OF,        |

|     |   |   |       |                                |          |
|-----|---|---|-------|--------------------------------|----------|
|     |   |   |       | NOUN, COERCE,<br>EXCEPTIVE_BUT | KIND_OF} |
| LOW | 7 | 4 | 0.095 | RECIP_EACH_OTHER,<br>FREE_REL  | —        |

## 6. Cross-linguistic mini-sets (uniform tests, uniform reporting)

This section expands the typology from illustrative to diagnostic. For each language we provide parallel minimal pairs aligned to DB tests, with judgment labels: [A] acceptable, [M] marginal, [C] context-dependent, [E] ellipsis-dependent. Examples are presented as reference-attested where possible; otherwise they are marked as 'template to be validated'.

### English: DB-G6 (closure ordering + PROX over Qpost)

**(E1)** Almost every student passed the exam.

**Status:** [A]

Context: Neutral classroom scenario; PROX modifies Qpost.

**(E2)** \*The every student passed the exam.

**Status:** [U] (predicted undefined)

Context: Intended D over post-closure Q; predicted type crash.

**(E3)** Each one passed the exam.

**Status:** [A] (repair: prop-noun ONE)

Context: Prop-noun repair supplies missing ⟨ e,t ⟩ when NP head is absent.

### English: DB-G5 (repair taxonomy: PRON vs REPAIR vs COERCE)

**(E4)** None left.

**Status:** [A] (PRON)

Context: Negative pronoun as lexical DP replacement.

**(E5)** \*No left.

**Status:** [U] (needs repair)

Context: Determiner no lacks NP head; requires repair (e.g., 'no one' or ellipsis).

**(E6)** The him was back again.

**Status:** [C] (COERCE)

Context: Typed coercion: persona/role predicate formed from pronoun.

**German: DB-G6 (PROX over Qpost; D over Qpost blocked)**

**(G1)** Fast jeder Student bestand die Pruefung.

*FAST every student passed the exam*

**Status:** [A] (reference-attested)

Context: Approximation/adverbial modifier over universal quantifier.

**(G2)** \*Der jeder Student bestand die Pruefung.

*the.NOM every student passed the exam*

**Status:** [U] (predicted)

Context: Attempted article over jeder; expected stacking failure.

**(G3)** Jeder einzelne Student bestand die Pruefung.

*every single student passed the exam*

**Status:** [A]

Context: Repair/intensification via adjective/packaging; remains within permissible pipeline.

**German: DB-G5 (negative repair: pronoun vs determiner)**

**(G4)** Keiner ist gekommen.

*no.one is come.PTCP*

**Status:** [A] (PRON)

Context: Negative pronoun as DP replacement.

**(G5)** Kein Student ist gekommen.

*no student is come.PTCP*

**Status:** [A] (NP supplied)

Context: Determiner + overt NP head; no repair needed.

**Mandarin: DB-G5/6 (mei + classifier; dou interaction; PROX)**

**(M1)** (Jihu) mei-ge xuesheng dou tongguo le kaoshi.

*almost every-CL student DOU pass ASP exam*

**Status:** [A] (reference-attested)

Context: Classifier required; dou often couples with mei-NP for distributivity.

**(M2)** Mei san-ge xuesheng dou ...

*every three-CL student DOU ...*

**Status:** [C] (kappa/partition)

Context: Group distributive requires salient partition/cover into 3-sized cells.

**(M3)** Suoyou (de) xuesheng dou ...

*all DE students DOU ...*

**Status:** [A] (Mode-1 ALL strategy)

Context: Preclosure ALL strategy distinct from mei.

**Turkish: DB-G6 (PROX over Qpost; negative quantifier)**

**(T1)** Neredeyse her ogrenci sinavi gecti.

*almost every student exam.ACC passed*

**Status:** [A] (reference-attested)

Context: Proximity over universal quantification.

**(T2)** Neredeyse hicbir ogrenci gelmedi.

*almost no student come.NEG.PST*

**Status:** [A] (reference-attested)

Context: Proximity over negative quantifier/determiner.

**(T3)** Her biri geldi.

*every one.POSS came*

**Status:** [A] (repair: prop-noun)

**English: DB1 (ellipsis control: bare quantifier heads)**

**(E7)** Some left early.

**Status:** [A] (indefinite pronoun as DP)

Context: Bare SOME here is pronominal; not a determiner without NP.

**(E8)** No students left early.

**Status:** [A] (overt NP head)

Context: Determiner no with NP predicate supplied.

(E9) No left early.

**Status:** [U] (needs repair)

Context: No as determiner without NP: requires 'no one' or ellipsis of NP if licensed.

**German: DB1 (ellipsis vs true bare headability)**

(G6) Einige gingen frueh.

*some went early*

**Status:** [A] (pronoun)

Context: Einige is pronominal (DP), not determiner-only.

(G7) Kein(e) Studenten gingen frueh.

*no students went early*

**Status:** [A]

Context: Overt NP supplied.

(G8) \*Kein ging frueh.

*no went early*

**Status:** [U] (needs repair)

Context: Requires pronoun keiner / ellipsis not generally available in this form.

**Mandarin: DB1 (classifier repair as required NP supplier)**

(M4) Mei-ge ren dou lai le.

*every-CL person DOU come ASP*

**Status:** [A]

Context: Classifier supplies nominal measure/head for mei.

(M5) \*Mei dou lai le.

*every DOU come ASP*

**Status:** [U]

Context: Without classifier/nominal supplier, mei cannot head DP.

(M6) Mei yi-ge dou lai le.

*every one-CL DOU come ASP*

**Status:** [A] (prop-noun/measure repair)

Context: yi-ge functions as repair supplier analogous to ONE.

**Turkish: DB1 (repair suppliers for distributives)**

**(T4)** Her öğrenci geldi.

*every student came*

**Status:** [A]

Context: Overt NP head supplied.

**(T5)** \*Her geldi.

*every came*

**Status:** [U]

Context: Determiner-like her lacks NP head.

**(T6)** Her biri geldi.

*every one.POSS came*

**Status:** [A] (prop-noun repair)

Context: biri supplies nominal head; parallels English 'each one'.

**Spanish: DB-G6 (article language: PROX over Qpost; D over Qpost)**

**(S1)** Casi todos los estudiantes aprobaron.

*almost all the students passed*

**Status:** [A] (template)

Context: ALL strategy; preclosure-like.

**(S2)** Casi cada estudiante aprobo.

*almost each student passed*

**Status:** [A] (template)

Context: PROX over each.

**(S3)** \*El cada estudiante aprobo.

*the each student passed*



**Status:** [U] (predicted template)

Context: D over each predicted to fail unless fused.

### Russian: DB-G6 (article-less: covert D-closure diagnostics)

**(R1)** Pocti kazhdyj student sdal ekzamen.

*almost every student passed exam*

**Status:** [A] (template)

Context: PROX over universal.

**(R2)** \*etot kazhdyj student ...

*this every student ...*

**Status:** [U]/[C] (predicted)

Context: Demonstratives may occupy D-like head; predict failure for D over Qpost unless reanalyzed.

**(R3)** Vse studenty sdal-i ekzamen.

*all students pass-PST.PL exam*

**Status:** [A] (template)

Context: ALL strategy with bare plural.

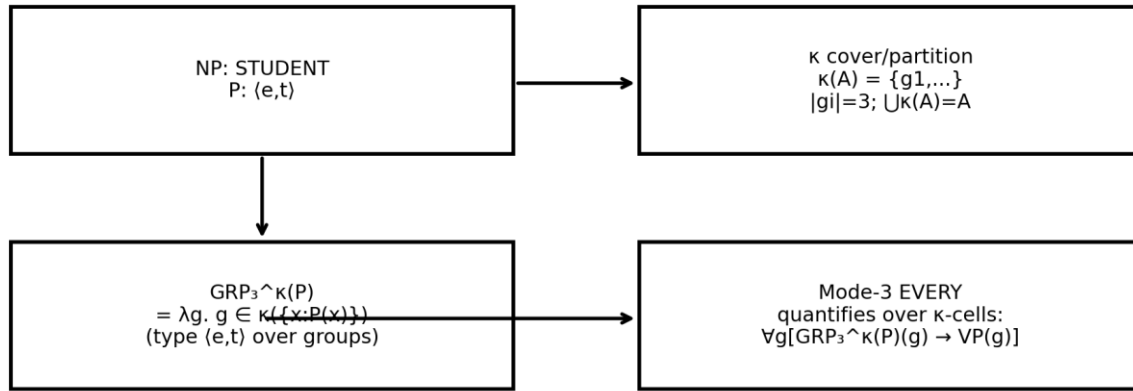
Context: Prop-noun repair for distributive universal.

## 6.5 Extension languages (templates for rapid expansion)

To broaden typological coverage in a thesis, add at least: (i) an article language with rich DP stacking diagnostics (e.g., Spanish), (ii) an article-less language (e.g., Russian) to test covert D-closure, and (iii) a classifier language beyond Mandarin (e.g., Japanese) to test classifier repair. The same DB tests apply; only the parameter grid changes (article realization, classifier availability, ellipsis/pro-drop).

## 7. A fully worked $\kappa$ -derivation (why structured $\kappa$ is required)

We now execute a typed  $\kappa$  derivation for group distributives such as 'every three students'. The goal is to make the  $\kappa$  requirement explicitly architectural: under HTQA Mode-3 postclosure traversal, distributivity requires addressable units. Numerals can shift the unit from atoms to groups, but the groups must be supplied as a structured domain.  $\kappa$  is that structured-domain parameter.



If  $\kappa$  is absent: group predicate undefined  $\rightarrow$  no traversal units

Figure 8.  $\kappa$ -licensed group distributive:  $\kappa$  supplies 3-sized cells; Mode-3 EVERY quantifies over  $\kappa$ -cells.

### 7.1 Typed steps

- Step 1 (NP):  $STUDENT := \lambda x.student(x)$  (type  $\langle e, t \rangle$ ).
- Step 2 ( $\kappa$ -supplied grouping):  $GRP_3^{\kappa}(STUDENT) := \lambda g. g \in \kappa(\{x : student(x)\})$  (type  $\langle e, t \rangle$  over group individuals).
- Step 3 (Mode-3 EVERY):  $EVERY\_Mode3$  takes an addressable predicate over units  $u$  (atoms or groups):  $\forall u[GRP_3^{\kappa}(STUDENT)(u) \rightarrow VP(u)]$ .
- Step 4 (Failure point): without  $\kappa$ , Step 2 is undefined; there is no predicate of addressable units for Mode-3 traversal.
- Step 5 (MIDDLE rescue options): supply  $\kappa$  via an overt grouping frame ('in groups of three'), via classifier/measure insertion (classifier languages), or via discourse-salient partition; otherwise the group reading is degraded or coerced.

### 7.2 Why this is 'DP/GQ-expensive'

A baseline generalized-quantifier determiner analysis can type-check 'every three students' by allowing numeral phrases to denote predicates over pluralities. However, it does not derive the  $\kappa$  licensing condition from general principles: it typically must stipulate a cover/partition requirement as a constructional constraint. In the tier-nested HTQA analysis,  $\kappa$  is demanded by entry-mode addressability: Mode-3 postclosure traversal quantifies over units that must be supplied as a structured domain.

## 8. Distinguishing predictions (tier containment + entry mode)

### P1. Entry-mode invariance (no 'MIDDLE changes HIGH mode')

Packaging/repair can supply missing inputs ( $\langle e, t \rangle$  suppliers,  $\kappa$  suppliers) but cannot convert a Mode-1 ALL derivation into Mode-3 postclosure traversal. Therefore, addressability phenomena that require Mode-3 cannot be produced merely by degree/partitive packaging over ALL.

### P2. Repair-locality implicational constraint

If a language lacks NP-ellipsis and lacks lexical quantifier-pronouns, then postclosure operators must co-occur with an overt MIDDLE supplier whenever the NP predicate is absent. The only permitted suppliers are the T3 set (ELL/PRON/REPAIR/COERCE); 'lexical idiosyncrasy' is replaced by a completeness-licensed necessity.

### P3. Fusion signature prediction

Implementation sketch: given a dependency-parsed corpus, collect candidate pairs (modifier, head) where the modifier is adjacent and attaches by an adjectival/adverbial/functional relation. Then apply a strict filter that rejects pairs where the modifier scopes over a phrase boundary (e.g., clausal adverbs) or where the target is not a head (e.g., determiner inside DP but modifier attached to NP). The resulting adjacency-conditioned graph can be compared to the hand-coded Tier-1 graph. A simple metric is F1 over edges, and a higher-level metric is whether the induced isolate profile matches the theoretical tier signature.

Any apparent D-over-postclosure-Q pattern must be morphological fusion (one HIGH head), not true head-head stacking. Independent modification of the subparts should be blocked, and the HIGH-tier graph should still show isolates under the strict criterion.

## 9. Computational instantiation (chapter-level plan)

The tier graphs can be induced from parsed corpora, but only if the strictness rubric is enforced. A practical computational chapter contribution is a two-stage filter: (i) harvest candidate modifier-head pairs from constituency or enhanced dependencies; (ii) reject candidates that fail strict head targeting or are scope-only. The output is a tier graph per language whose isolate set and SCCs can be compared across genres. A minimal evaluation is an ablation: removing the strict head-targeting filter should collapse the predicted HIGH-tier isolate profile by flooding the graph with scope edges, thereby validating the need for the strict criterion.

## 10. Neuroscience mapping (chapter-level predictions)

The tier containment hypothesis predicts a processing dissociation between closure (HIGH selection/closure) and repair (MIDDLE supplier recruitment). In comprehension, encountering a postclosure operator without an available  $\langle e, t \rangle$  supplier should trigger repair: ellipsis reconstruction, pronounization, prop-noun insertion, or coercion. These repairs should produce measurable signatures (reading-time increases, ERP/MEG reanalysis components) distinct from

ordinary scope composition. A thesis-ready experiment manipulates (i) postclosure presence, (ii) overt NP availability, and (iii) repair strategy, while holding propositional content constant.

### Appendix A. Edge decision rubric (strict head-modification)

1. A1 Head targeting: verify X attaches to Y as head (not merely to YP).
2. A2 Constituency: verify [X Y] forms a constituent excluding the rest.
3. A3 No category shift: exclude quotation, metalinguistic mention, or POS conversion.
4. A4 Scope exclusion: if X scopes over phrase containing Y without targeting Y as head, do not add edge.
5. A5 Ellipsis/pro-drop control: if Y can be silent NP, label ellipsis-supported (do not count as bare headability).
6. A6 Repair coding: classify rescue as ELL vs PRON vs REPAIR vs COERCE.

### Appendix B. Results reporting template (ready for elicitation or reference-attested coding)

| DB test | Prediction | Language | Evidence tier | N_speakers | N_items | Agreement (%) | Notes / variance / source conflict |
|---------|------------|----------|---------------|------------|---------|---------------|------------------------------------|
|---------|------------|----------|---------------|------------|---------|---------------|------------------------------------|

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### **Appendix C. Instantiated edge lists (Tier-1 coding)**

HIGH tier edges: none (all nodes are isolates under strict head-modification).

MIDDLE tier edges ( $X \rightarrow Y$ ): OTHER  $\rightarrow$  ONE; KIND\_OF  $\rightarrow$  SORT\_OF; SORT\_OF  $\rightarrow$  KIND\_OF; REPAIR\_FRAME  $\rightarrow$  PARTITIVE\_OF.

MIDDLE tier SCCs: {KIND\_OF, SORT\_OF}.

LOW tier edges ( $X \rightarrow Y$ ): RELCL\_MOD  $\rightarrow$  PERSONAL\_PRONOUN;  
RELCL\_MOD  $\rightarrow$  INDEF\_PRONOUN; RELCL\_MOD  $\rightarrow$  WH\_HEAD; (optionally)  
LEX\_NP  $\rightarrow$  LEX\_NP for noun-noun compounding (lexical modifier).